

Directional Causality and the Occlusion Principle: A Null-Shell Framework for Electromagnetic and Gravitational Interaction Strength

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Abstract This paper develops the occlusion principle: the proposal that interaction strengths in electromagnetic and gravitational systems are governed not by local field amplitudes but by the availability of null directions through which causal contact can occur. Field superposition eliminates amplitude while preserving the full directional occupancy of null rays. Dense environments—from charged cavities to galactic cores—can therefore saturate the causal channels required for interaction, reducing effective coupling strengths even when classical theory predicts zero field or weak fields. We show how this perspective reframes Debye screening, Pauli blocking, tight photon–baryon coupling, the classical radiation reaction problem, and the correlation between baryonic density and dark-matter-like phenomena. We further analyze the uniformly charged shell and Olbers-type configurations as diagnostic systems in which field-based and directional-causality frameworks make distinct, falsifiable predictions.

1 Introduction

The superposition principle is one of the most powerful tools in classical physics. It allows electromagnetic and gravitational fields from multiple sources to be summed algebraically, producing compact analytic results in situations ranging from charged cavities to galactic mass distributions. Yet superposition carries a conceptual cost: in computing the net field amplitude, it discards the underlying directional structure of the contributing influences. Two strong but oppositely directed signals cancel to zero not because they cease to exist, but because the mathematical representation of the field retains only their sum.

This distinction is normally harmless. However, several well-studied physical phenomena already hint that interaction strength depends on more than field amplitude alone. Debye screening reduces Coulomb interactions in plasmas even though the local field vanishes only in a mean-field sense [1, 2]. Pauli blocking suppresses electromagnetic transitions in dense fermionic systems despite no reduction

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in classical field strength [3]. Tight photon–baryon coupling in the early universe operates through directional constraints rather than field cancellation [4, 5]. Even the classical radiation-reaction problem reveals sensitivity to the structure of null propagation rather than local field values [6, 7].

These observations suggest that the field formalism may be incomplete as a description of the causal pathways through which interactions occur. In particular, they raise a question that is largely absent from the classical framework:

Does interaction strength depend on the availability of null directions between particles, rather than solely on the net field amplitude at their locations?

This paper develops the *occlusion principle*, the proposal that electromagnetic and gravitational interactions are mediated by null-shell contact: causal influence propagating along lightlike intervals. In this view, interaction strength is determined by whether two particles can access null directions that remain preferentially available relative to those saturated by the surrounding environment. Superposition can erase field amplitude but cannot restore null directions that have been saturated by surrounding matter. As a result, environments that appear “field-free” in the classical sense may nevertheless suppress interactions by exhausting the available causal channels.

The uniformly charged shell and Olbers-type configurations serve as clean diagnostics of this distinction. Both systems exhibit inverse-square cancellation in the field description and appear dynamically inert. Yet from the perspective of directional causality, they represent maximally saturated environments in which few or no null directions remain unoccupied. This leads to experimentally testable differences in interaction strength and offers a unified perspective on phenomena typically treated as unrelated: screening in plasmas, suppressed transitions in dense fermionic media, cosmological opacity, and density–acceleration correlations often attributed to dark matter [9, 10].

The goal of this work is not to modify Maxwell’s equations or Newtonian gravity, but to clarify the causal structure implicit in their solutions and to identify when the field description may be insufficient to characterize the physical environment. The occlusion principle provides a complementary interpretation whose predictions differ precisely when directional saturation becomes significant.

1.1 The problem hidden inside superposition

Superposition is an amplitude-level operation. When two fields $\mathbf{E}_1$ and $\mathbf{E}_2$ sum to zero at a point, their vectorial contributions have opposite direction and equal magnitude. The underlying null rays that carried these influences are still present in the environment, but the field description no longer encodes them. Thus the field can be zero even while the causal environment is densely populated with incoming influence.

This blind spot is rarely consequential in dilute settings where directional occupancy is low. But as matter density increases or geometric symmetry forces all null directions to intersect surrounding matter, superposition ceases to track the physical variable that determines interaction availability.

1.2 Why a zero field does not imply zero influence

A zero field traditionally implies that a test particle experiences no net force. This is correct under the assumption that interaction strength depends only on the net field amplitude at its location. However, if interaction strength depends on the *differential* availability of null directions between particles, then the absence of a net field does not guarantee that the directional contrast required for a force is present. In symmetric environments such as spherical shells or homogeneous cosmological distributions, every direction from a point can be continuously filled by influence from elsewhere. In that case, the field may vanish while the causal environment is maximally active and directional contrast is eliminated.

1.3 Existing phenomena implying directional saturation

Several physical processes are naturally interpreted as consequences of directional saturation: plasma screening limits null access through collective charge response, Pauli blocking restricts available final states for electromagnetic transitions, tight photon–baryon coupling arises from repeated null-contact events in a dense environment, and radiation reaction in accelerated systems emerges as an asymmetry in null-shell occupancy. These diverse phenomena suggest that directional constraints—not just field amplitudes—play a determining role in the strength of interactions [1, 2, 3, 4, 6, 7].

1.4 Overview of the occlusion principle

The occlusion principle asserts that available null directions form a limited resource. When an environment fills these directions with incoming influence, particles within it lose the ability to form direct causal contact with one another. The classical field description, which tracks amplitude but not occupancy, cannot register this saturation. The remainder of the paper develops this idea formally, applies it to symmetric environments in which the difference between amplitude and directional occupancy is stark, and identifies empirical situations in which the two frameworks make distinct predictions.

2 The Occlusion Principle

The occlusion principle is based on a single operational hypothesis: electromagnetic and gravitational interactions occur through lightlike causal contact between particles. This is consistent with the long-standing use of retarded and advanced Green’s functions in classical electromagnetism and with time-symmetric formulations such as the Wheeler–Feynman absorber theory [7]. Here we treat null-shell contact not as a mathematical device but as the physical mechanism underlying interaction.

2.1 Null-shell contact as the mechanism of interaction

Let two particles at positions x_1 and x_2 interact. In a null-contact framework, causal influence between them propagates along lightlike intervals. The geometric null channel linking the particles always exists; what determines the strength of the interaction is not its existence but its *relative availability* compared to other null directions.

Environmental matter populates null directions with its own outgoing influence. When this occupancy is anisotropic, the directional channel toward another particle remains privileged, producing a net force. When occupancy is uniform, this directional contrast is erased, and the interaction weakens even though the geometric null channel remains present.

2.2 Directional availability vs. field amplitude

Field amplitude does not track the *differential* availability of null channels. Two fields that cancel to zero may nonetheless saturate all null directions equally, eliminating the directional contrast required to generate a force. Conversely, a nonzero field may coexist with abundant differential availability.

The occlusion principle therefore states:

Interaction strength is proportional to the *differential* availability of null directions pointing toward another particle relative to those saturated by the environment.

It is this contrast—not the mere presence or absence of a geometric null path—that produces a measurable force.

2.3 Saturation and blockage of null directions

A null direction is *saturated* if it is continuously occupied by incoming influence from surrounding matter. Saturation reduces interaction strength not by removing the null path between particles, but by reducing the *relative* availability of that direction compared to its neighbors.

In the limit where every direction is equally saturated, the contrast between “toward the other particle” and “toward the rest of the environment” vanishes, and the effective interaction tends to zero.

2.4 Why fields erase directional information

The mathematical structure of classical fields forces a projection from the full set of directional contributions to a single three-vector at each point. This projection collapses directionally distinct influences into a single amplitude. When many such influences cancel, the resulting field contains no record of the original directional occupancy. As a consequence, superposition hides precisely the information that determines interaction strength in the occlusion framework.

The remainder of the paper identifies environments in which this hidden variable becomes operationally significant and explores the empirical implications.

3 Empirical Phenomena Consistent With Occlusion

The occlusion principle is not introduced in isolation; several well-known physical processes already behave as though interaction strength is limited by directional availability rather than field amplitude. These phenomena are usually treated as unrelated and explained within different subfields. Here we highlight how each finds a unified interpretation when interactions are viewed as null-shell contact processes whose strength depends on the availability of unoccupied null directions.

3.1 Debye screening

In a plasma, mobile charges rearrange to reduce the effective Coulomb interaction over distances larger than the Debye length. The standard explanation invokes collective response and partial field cancellation [1, 2]. From the perspective of directional causality, Debye screening is equivalently understood as a geometric saturation effect: a test charge is surrounded by a dense distribution of particles whose null emissions occupy an increasingly large fraction of the available directions. The test charge’s ability to form clean null channels with distant charges is consequently diminished. Thus the reduction in effective interaction is not caused solely by field cancellation but also by the saturation of causal pathways.

3.2 Pauli blocking and phase-space occlusion

In degenerate fermionic systems, many electromagnetic transitions are forbidden even when the classical field strength is unchanged. Traditionally this is attributed to the exclusion principle restricting final states. Within the occlusion framework, Pauli blocking appears as a directional effect: fermions in filled momentum states continuously populate specific null directions with their emissions, leaving fewer open channels for additional interactions. The suppression of radiative processes is again linked to a loss of available null contact paths, not merely to amplitude considerations [3].

3.3 Photon–baryon coupling in the early universe

Before recombination, photons and baryons behaved as a tightly coupled fluid. This coupling persisted despite photons experiencing very large mean free paths compared with laboratory plasmas. The standard explanation relies on repeated Thomson scattering and baryon acoustic oscillations [4, 5]. Directional causality provides a complementary view: the immense baryon density filled nearly all null directions with ongoing interactions, limiting the availability of clean channels through which photons could propagate freely. The effectiveness of the coupling followed from directional saturation, not the instantaneous field strength at any point.

3.4 Refractive index and optical opacity

A refractive index greater than unity indicates that electromagnetic waves experience repeated microscopic interactions as they propagate through a medium. From an occlusion standpoint, the medium occupies a fraction of the null directions that would otherwise be available for unimpeded propagation. The opacity of dense materials corresponds to the near-complete saturation of those directions. Field amplitude plays no essential role in this description; what matters is the occupancy of null channels by the material constituents.

3.5 Radiation reaction as null-shell asymmetry

Classical radiation reaction is usually presented as a self-interaction problem that requires renormalization or advanced potentials [6]. However, genuine self-interaction is incompatible with relativity: a particle cannot exert an influence on itself across a null interval, because a null-separated “self” has zero spatial separation and no distinct causal location. Any formulation that treats the particle as interacting with a field evaluated at its own instantaneous position therefore violates the basic causal structure of spacetime.

Time-symmetric approaches such as Wheeler–Feynman [7] avoid this contradiction by interpreting radiation reaction as arising from interactions with the absorber—the distribution of other charged matter in the universe.

In the occlusion framework, the effect appears naturally. An accelerating charge distorts the symmetry of its past and future null-shell contact regions, producing an imbalance in directional occupancy. The resulting momentum recoil is proportional to $\dot{a}$, matching the form of the classical radiation-reaction term. No self-interaction is required; the effect follows from the causal geometry of null directions.

3.6 Density–dark-matter correlation

Observationally, galaxies with higher baryonic densities exhibit stronger deviations from Newtonian rotation curves [9, 10]. This scaling is difficult to reconcile with particle dark matter models but follows immediately from the occlusion principle. In dense galactic cores, baryonic matter saturates many of the null directions required for gravitational contact, reducing the effective coupling between masses. Outer regions, with far lower density, experience less saturation and therefore higher effective coupling. The resulting gradient in directional availability produces the same qualitative signatures attributed to dark matter, without introducing new particles or modifying the inverse-square law.

These examples demonstrate that directional saturation is already implicit in a broad range of physical processes. The occlusion principle provides a unifying conceptual framework that connects them through a single causal mechanism.

4 The Uniform Shell and Olbers-Type Distributions as Diagnostics

A central question for any proposed modification or reinterpretation of electromagnetic or gravitational interaction is whether the framework produces distinct,

experimentally testable deviations from classical expectations. The occlusion principle does not modify Maxwell’s equations or the inverse-square law; rather, it reinterprets interaction strength as being limited by the availability of null directions along which causal contact may be formed. Because field superposition encodes only amplitude—not the directional occupancy of the underlying null rays—there exist physical configurations in which the field-based and directional-causality descriptions make divergent operational predictions. The uniformly charged (or massive) shell provides the simplest such environment.

4.1 Interior field cancellation and classical expectations

Classical electrodynamics and Newtonian gravity both assert that the field inside a uniformly charged or massive spherical shell is exactly zero. By the usual interpretation, this cancellation implies that the interior region is physically indistinguishable from free space: any two charges or masses placed inside the shell should interact with precisely the same coupling strength as they would in vacuum. This conclusion is internally consistent within the field framework, because the inverse-square decrease in strength from each surface element is exactly offset by the increase in available surface area, yielding a vanishing net field.

4.2 Null-direction saturation inside a symmetric shell

The occlusion principle asks a different question from classical field theory. Instead of evaluating the net field amplitude at a point, it examines the availability of null directions required for causal contact. Every surface element of the shell continuously emits electromagnetic or gravitational influence along null rays. Although these contributions cancel in the sense of net field amplitude, they do not eliminate the underlying directional occupancy.

From any interior point, every null direction intersects some region of the surrounding shell. Geometrically, two particles inside the shell still possess direct null-separated paths between them; the causal link exists. What changes is the *differential* availability of null directions. If the shell saturates all directions uniformly, then the directional contrast between “toward the other particle” and “toward the rest of the environment” vanishes.

Since interaction strength in the occlusion framework is determined by differences in directional availability rather than by the mere existence of geometric null paths, a perfectly symmetric shell drives this differential toward zero. The interior becomes a maximally saturated causal environment in which no direction is more available than any other.

4.3 Interaction suppression under the occlusion principle

Within the occlusion framework, two charges or two masses inside a uniform shell experience reduced—and, in the limit of perfect symmetry, vanishing—effective coupling. This does not occur because the geometric null channel between the particles disappears; it does not. Rather, the null direction connecting the particles

becomes no more available than any other direction, because every direction is already continuously occupied by influence arriving from the shell.

Interaction requires directional *asymmetry*. In free space, the null-channel region pointing toward another particle is uniquely available; the surrounding environment does not impose symmetric saturation. Inside a perfectly symmetric shell, that asymmetry is erased. The directional contrast that produces a net force collapses to zero.

Thus, under ideal symmetry: *two freely falling bodies inside a massive shell should exhibit no net gravitational attraction*, not because the link between them is lost but because it is no longer privileged relative to any other direction. The same reasoning applies to electromagnetic coupling inside a charged shell.

This effect is conceptually analogous to Pauli blocking or Debye screening—not through exclusion or charge redistribution, but through the elimination of directional contrast. Even though the classical field is zero, the causal environment is densely occupied, and the conditions for a net interaction are absent.

It is also noteworthy that general relativity predicts nonzero gravitational time dilation inside a massive spherical shell: clocks inside the cavity run at a different rate relative to a distant observer, despite the classical statement that the gravitational “field” is zero. This reinforces the central point of the occlusion framework. Field cancellation does not imply causal emptiness. The gravitational potential and the directional structure that governs causal propagation remain present even when the net force vanishes, and time dilation provides a direct operational signature of this remaining causal structure.

4.4 Connection to Olbers’ paradox and inverse-square cancellation

The uniformly charged shell is not the only system in which inverse-square cancellation obscures the underlying causal structure. Olbers’ paradox provides an instructive parallel [8]. In an infinite, static, homogeneous universe, the $1/r^2$ decrease in flux from distant stars is exactly offset by the r^2 growth in the number of sources, implying that every line of sight should terminate on a luminous surface. Modern cosmology resolves this paradox via the finiteness of the observable universe and the cosmic expansion, not by altering the inverse-square law.

The shell theorem is a finite version of the same geometric cancellation: the $1/r^2$ decrease in contribution from each surface element is offset by the growth of the surrounding surface area. In both cases, the field framework treats the cancellations as implying a low-influence or zero-influence environment.

Directional causality reaches a different conclusion. Inverse-square cancellation does not imply emptiness; it implies that *every direction is filled*. An infinite homogeneous distribution saturates all lines of sight. A uniform spherical shell saturates all null directions from any interior point. In both cases, the causal environment is fully occupied even when the amplitude-based field description assigns zero or finite values.

4.5 Why field theory and directional causality diverge here

The divergence between the field and occlusion descriptions arises because superposition discards the very information that determines interaction availability in the occlusion framework. A zero field is interpreted classically as the absence of influence, while the occlusion principle interprets the same condition as maximal directional occupancy. Thus the uniformly charged shell is not a contradiction of Maxwell’s theory; rather, it exposes a variable—the availability of null directions—that Maxwell’s equations do not track. The system therefore serves as a clean diagnostic: if interactions between interior particles are unchanged, the field description is sufficient. If they are suppressed, directional causality is required.

4.6 Experimental and observational discriminators

The uniformly charged shell provides a tabletop-scale test of the occlusion principle. If null-direction saturation governs interaction strength, then two charges placed inside a conductive or high-voltage cavity should exhibit a measurable reduction in effective Coulomb coupling relative to their behavior in free space. Analogous gravitational tests are less accessible in the laboratory but appear naturally in astrophysical systems. Galactic cores, where baryonic density is highest, should display reduced effective gravitational coupling—precisely the environment in which dark-matter-like deviations are typically inferred. The symmetry between small-scale and large-scale saturation effects is a central empirical prediction of the occlusion framework.

5 Coupling Constants as Directional Fractions

If interaction strength depends on the availability of unoccupied null directions rather than on local field amplitude, then the fundamental constants associated with electromagnetic and gravitational coupling acquire a new interpretation. In the standard formulation, the fine-structure constant α and Newton’s gravitational constant G are universal parameters whose values are fixed inputs to the theory. In a directional-causality framework, these constants take on a geometric meaning: they represent the fraction of null directions that are not saturated by the surrounding matter distribution.

5.1 Reinterpreting α as an occlusion fraction

The fine-structure constant,

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}, \quad (1)$$

is dimensionless and historically regarded as a fundamental property of nature. Yet numerous effective-field calculations require environment-dependent modifications of α , and condensed-matter analogues routinely exhibit different apparent couplings. In the occlusion picture, such variations arise naturally: α is interpreted

as the fraction of null channels between two charges that remain unoccupied by the rest of the environment.

Let f_{EM} denote the fraction of available null directions after accounting for background occupancy. Then the effective electromagnetic coupling is

$$\alpha_{\text{eff}} = f_{\text{EM}} \alpha_0, \quad (2)$$

where α_0 is the vacuum value corresponding to full directional availability. Dense or highly structured environments reduce f_{EM} by filling null directions with ongoing electromagnetic influence, thereby suppressing effective coupling without altering the microscopic properties of the charges.

Importantly, f_{EM} does not measure the absolute number of unsaturated null directions but the *relative contrast* between directions pointing toward another charge and those occupied by the environment. Interaction requires anisotropy in the null-shell availability; uniform saturation suppresses this anisotropy even if the geometric null channel between charges remains open.

5.2 Reinterpreting G as a cosmological-direction fraction

Gravitational interaction can be treated analogously. If causal contact occurs along null geodesics and interaction strength is proportional to the number of available null channels, then G encodes the fraction of gravitational null directions that remain unsaturated by the global matter distribution.

Let f_{grav} be the gravitational directional-availability fraction. Then

$$G_{\text{eff}} = f_{\text{grav}} G_0, \quad (3)$$

where G_0 is the value corresponding to an environment in which no gravitational null directions are occluded. In this interpretation, the smallness of the observed gravitational coupling does not reflect a fundamental disparity in the underlying laws, but the fact that only a limited fraction of all null directions contributes effectively to gravitational contact once the global matter distribution is taken into account.

This viewpoint immediately connects gravitational anomalies to matter density. Regions of high baryonic density suppress f_{grav} , yielding a smaller effective coupling. Regions of low density preserve more null directions and therefore exhibit stronger effective coupling. This density dependence mirrors the observed variation in galactic rotation curves traditionally attributed to dark matter [9, 10].

Thus G_{eff} reflects the degree to which the null directions linking two masses remain preferentially available relative to those filled by the cosmic matter distribution. In regions of high symmetry and high density, this contrast collapses, suppressing effective gravitational coupling.

5.3 Matter distribution as a regulator of effective couplings

Both f_{EM} and f_{grav} depend on the local and cosmological distribution of matter. In homogeneous environments, the occupancy of null directions follows directly from geometric considerations: the number of sources contributing to directional

saturation grows with surface area, while the null intensity from each source falls as $1/r^2$, producing patterns of cancellation similar to those in the shell theorem and Olbers-type arguments. In dense, irregular, or multi-component environments, the occupancy is anisotropic and can vary over short length scales.

This framework provides a natural explanation for why effective couplings in materials, plasmas, galactic cores, and cosmological media differ from their vacuum values. No modification to the underlying inverse-square law is required; the variation arises from geometric saturation.

5.4 Emergent constants vs. microscopic inputs

In the occlusion framework, the observed values of α and G are constants in the operational sense: they describe the effective coupling strengths measured in the present-day universe. What the framework revises is not their constancy but their interpretation. Rather than representing fixed microscopic parameters, both constants encode the directional availability of null channels in the environment in which they are measured.

Because directional availability depends on the large-scale matter distribution, the effective values of α and G need not be the same in all epochs. During the early universe, when densities and symmetries differed significantly from today, the null-direction structure would also have been different. In that case, both α_{eff} and G_{eff} may have taken values distinct from those observed now, with the modern “constants” emerging only as the universe diluted and the directional environment stabilized.

This perspective therefore distinguishes between:

- the present-day values of α and G , which appear constant because the directional environment of the late universe is stable, and
- their underlying causal roles, which allow for different effective values whenever the null-direction geometry of spacetime changes, such as during the early high-density phase of cosmic history.

In this view, α and G remain constants of nature only in the limited sense that the current universe presents a fixed directional environment. Their deeper significance lies in measuring the openness of the causal pathways through which electromagnetic and gravitational interactions operate.

6 Cosmological Implications

If electromagnetic and gravitational interaction strengths are governed by the availability of unoccupied null directions, then large-scale cosmology provides a natural arena in which such directional effects become significant. Cosmic matter is not distributed uniformly; its density varies across many orders of magnitude. Consequently, the fraction of accessible null channels also varies. The occlusion principle predicts systematic trends in gravitational dynamics and radiative behavior that mirror several key observational features of galaxies and the early universe [9, 10, 4, 5].

In the occlusion framework, large-scale gravitational behavior depends not only on how many null directions are saturated but also on how *uniform* that saturation is. Perfectly symmetric saturation removes the directional contrast required for strong interaction, while asymmetric or dilute environments restore it. This contrast-driven interpretation underlies the cosmological trends described below.

6.1 Rotation curves from gravitational occlusion

Disk galaxies exhibit approximately flat rotation curves, indicating orbital velocities that exceed Newtonian expectations based on the observed baryonic mass. This discrepancy is traditionally attributed either to particle dark matter or to modifications of gravity at low acceleration scales [10,9].

Under the occlusion principle, no new particles or force laws are required. In dense galactic cores, matter saturates a larger fraction of gravitational null directions, reducing the effective coupling G_{eff} . As radius increases, baryonic density decreases sharply; fewer null directions are saturated, and G_{eff} increases toward its vacuum value. Thus the gravitational influence felt by orbiting stars becomes stronger with radius, counteracting the expected Keplerian decline.

This produces the qualitative rotation-curve behavior associated with dark matter, but arises from directional saturation rather than from additional mass or modified dynamics. The occlusion framework therefore provides a causal interpretation of the empirical “acceleration–baryon” relations observed across galaxies [9].

6.2 High-density galaxies and “dark matter amplification”

Galaxies with extremely high baryonic densities—starburst galaxies, massive ellipticals, and compact dwarfs—exhibit the strongest inferred dark-matter fractions. This trend is difficult to reconcile with particle dark matter, which is expected to cluster more weakly than baryonic matter, not more strongly.

Directional causality predicts the opposite correlation: greater baryonic density leads to stronger saturation of gravitational null directions and therefore to a larger suppression of effective coupling. Observers interpreting the suppressed gravitational response through a field-amplitude framework will infer additional unseen mass. This “dark matter amplification” is thus a diagnostic signature of occlusion rather than of exotic particles.

6.3 Early-universe saturation and inflation-like behavior

At very early times, the universe was vastly more dense than it is today. In such conditions, saturation of null directions would have been nearly complete. As a result, G_{eff} would have been significantly smaller than its present value. A suppressed effective gravitational coupling during early epochs reduces the decelerating influence of gravity on cosmological expansion. As the universe expands and density falls, f_{grav} —and thus G_{eff} —increase.

This produces an inflation-like behavior without requiring a scalar field, potential energy, or new degrees of freedom: the universe expands rapidly when G_{eff} is small, then transitions to slower expansion as gravitational null directions become increasingly available. This picture qualitatively resembles “fast-roll” and “variable- G ” cosmologies but emerges from a purely geometric causal mechanism [5].

6.4 Continuity between epochs of different occlusion levels

A key advantage of the occlusion framework is continuity. The same mechanism that reduces effective coupling in dense galactic cores also influences early-universe dynamics. No new physics is invoked at different scales; only the fraction of available null directions varies.

The progression may be summarized as follows:

- **Early universe:** nearly complete saturation of null directions; $G_{\text{eff}} \ll G_0$; rapid expansion.
- **Galactic halos and outer disks:** low baryonic density; little saturation; G_{eff} approaches its vacuum value.
- **Galactic cores:** moderate to high density; partial saturation; reduced G_{eff} and apparent “missing mass.”

This continuous variation provides a unified explanation for inflation-like early dynamics, mid-scale galactic behavior, and large-scale structure formation without appealing to new particles or violations of the inverse-square law.

7 Predictions and Tests

The occlusion principle differs from the classical field framework precisely when null-direction saturation becomes significant. Such environments arise naturally in laboratory settings, in astrophysical systems, and across cosmological epochs. This section identifies concrete tests that discriminate between the two frameworks without requiring modifications to Maxwell’s equations, general relativity, or the inverse-square law.

7.1 Laboratory systems: charged shells, cavities, and plasmas

The simplest diagnostic is the uniformly charged or conducting shell. Classical theory predicts that two test charges inside such a cavity experience the same Coulomb interaction as they would in free space. The occlusion principle predicts reduced coupling due to saturation of null directions by the shell.

A feasible experiment would place two ions or trapped electrons inside a conductive cavity whose interior field is classically zero. Measuring the equilibrium separation or oscillation frequency of the pair as a function of cavity charge or geometry can reveal whether the effective Coulomb constant is suppressed. A *null result* favors the classical view; measurable suppression supports directional causality.

Plasmas provide a second diagnostic. Debye screening is well understood, but the occlusion framework predicts that screening should vary with directional anisotropy even when scalar density remains fixed [1, 2]. Tunable anisotropic plasmas could reveal such directional dependence.

7.2 Astrophysical systems: density-dependent G_{eff}

Galactic rotation curves already suggest that gravitational coupling varies with baryonic density, though this is traditionally interpreted through the lens of dark matter [10, 9]. If occlusion governs gravitational contact, then galaxies with similar baryonic distributions should exhibit similar deviations from Newtonian dynamics regardless of their dark-matter content. Conversely, galaxies with atypically high or low densities should show predictable variations in inferred G_{eff} .

Predictions include:

- high-density ellipticals should show *stronger* apparent dark-matter fractions than low-density disks of the same stellar mass,
- ultra-diffuse galaxies should exhibit minimal deviations from Newtonian dynamics, and
- the characteristic acceleration scale extracted from rotation curves should correlate with baryonic surface density rather than with halo mass [9].

These trends are distinct from those predicted by particle dark matter and by modified-gravity frameworks.

7.3 Cosmological variation in effective couplings

If f_{grav} increases as cosmic density decreases, then G_{eff} should evolve gradually over cosmological time. Such evolution would impact:

- growth of large-scale structure,
- baryon acoustic oscillation peaks,
- cosmic chronometer measurements,
- and the inferred equation of state of dark energy [5].

Current observations constrain large variations but allow small, density-dependent drift in effective coupling. Precision cosmological surveys may be able to distinguish the occlusion prediction—a smooth, monotonic evolution tied to matter density—from other models with very different signatures.

7.4 Distinguishing occlusion from modified gravity and particle dark matter

Modified-gravity models alter the force law, while particle dark matter adds additional mass-energy. The occlusion framework keeps the inverse-square law intact and does not introduce new degrees of freedom. Instead, it predicts:

- reduced effective coupling in regions of high baryonic density,
- increased effective coupling in low-density regions,
- and a continuous density-driven transition between these regimes.

These predictions differ fundamentally from approaches in which the force law depends on acceleration, curvature, or hidden mass components. Astrophysical and laboratory tests therefore provide sharp discriminators between the frameworks.

8 Conclusion

This work develops the occlusion principle: the proposal that electromagnetic and gravitational interactions are governed by the availability of unoccupied null directions rather than by local field amplitudes. Field superposition hides directional structure, and a zero field does not imply that the causal environment is empty. In many environments—from plasmas and degenerate matter to galactic cores and early cosmological conditions—null directions can become partially or fully saturated by the surrounding matter distribution.

The occlusion framework reinterprets several otherwise disparate phenomena through a single causal mechanism: Debye screening, Pauli blocking, tight photon–baryon coupling, radiation-reaction asymmetry, and density–acceleration correlations traditionally attributed to dark matter all emerge from variations in directional availability rather than from modifications of underlying laws [1, 2, 3, 4, 6, 7, 9, 10]. The uniformly charged shell and Olbers-type configurations serve as diagnostics in which the difference between amplitude-based and direction-based interpretations becomes particularly clear [8].

A key implication is that the coupling constants α and G take on a geometric meaning: they measure the fraction of null directions that remain unoccupied in a given environment. Their effective values depend on matter density and distribution rather than on new particles or modified dynamics. This provides a unified explanation for suppressed interactions in dense media, for flat rotation curves in galaxies, and for the inflation-like behavior of the early universe.

The occlusion principle is falsifiable. Laboratory experiments using charged cavities and anisotropic plasmas, as well as astrophysical and cosmological surveys, can distinguish its predictions from those of classical field theory, particle dark matter, and modified gravity [5]. No modification to Maxwell’s equations or to the inverse-square law is required; rather, the causal interpretation of those laws is refined.

Future work will focus on quantitative models of directional occupancy, numerical simulations of null-channel saturation, and derivations of effective couplings in structured environments. If the occlusion framework is correct, the apparent complexity of electromagnetic and gravitational phenomena may be traced to a single, simple principle: interaction strengths measure the openness of the causal pathways through which particles communicate.

Data Availability Statement

This research does not use experimental data or numerical datasets. All arguments and conclusions are derived analytically within the manuscript. No external data were generated or analyzed in this study.

References

1. D. R. Nicholson, *Introduction to Plasma Theory* (Wiley, New York, 1983).
2. C. Hansen, A. B. Reimann, and J. Fajans, *Phys. Plasmas* **3**, 1820 (1996). doi:10.1063/1.871685
3. Y. Margalit, Y.-K. Lu, F. C. Top, and W. Ketterle, *Science* **374**, 976 (2021). doi:10.1126/science.abi6153
4. P. J. E. Peebles and J. T. Yu, *Astrophys. J.* **162**, 815 (1970). doi:10.1086/150713
5. D. H. Weinberg, M. J. Mortonson, D. J. Eisenstein, C. Hirata, A. G. Riess, and E. Rozo, *Phys. Rep.* **530**, 87 (2013). doi:10.1016/j.physrep.2013.05.001
6. P. A. M. Dirac, *Proc. R. Soc. Lond. A* **167**, 148 (1938). doi:10.1098/rspa.1938.0124
7. J. A. Wheeler and R. P. Feynman, *Rev. Mod. Phys.* **17**, 157 (1945). doi:10.1103/RevModPhys.17.157
8. P. S. Wesson, *J. Br. Astron. Assoc.* **99**, 10 (1989).
9. S. S. McGaugh, F. Lelli, and J. M. Schombert, *Phys. Rev. Lett.* **117**, 201101 (2016). doi:10.1103/PhysRevLett.117.201101
10. M. Milgrom, *Astrophys. J.* **270**, 365 (1983). doi:10.1086/161130